GRAPH THEORY

Lecture 8

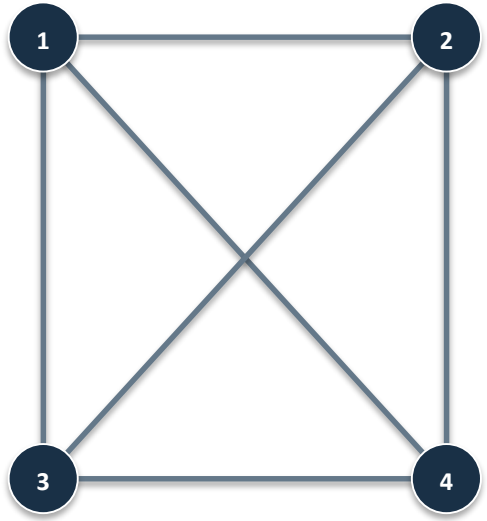
Planarity: Euler Bounds to Kuratowski Obstructions

Embeddings • Face Counting • Edge Bounds • Subdivision • Homeomorphism

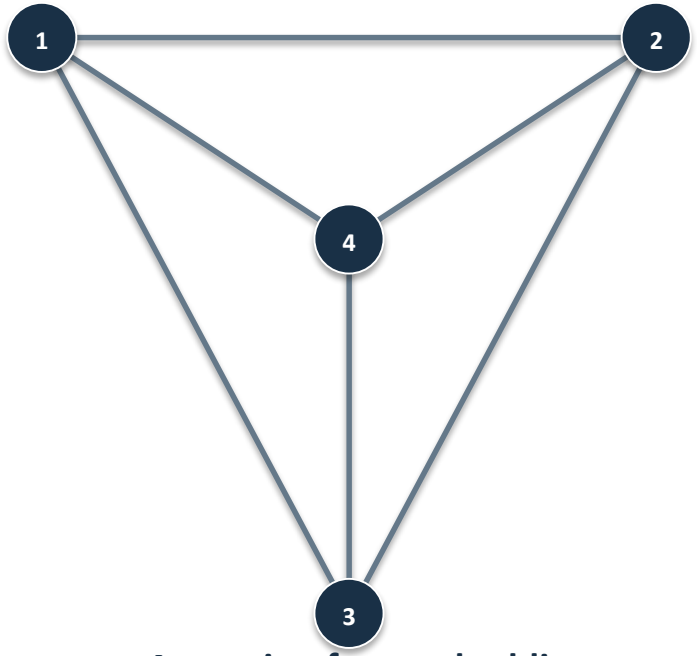
Numerical obstruction → Structural obstruction

From a Drawing to a Plane Embedding

Quick recap — the graph property is independent of a particular drawing



A drawing with crossings

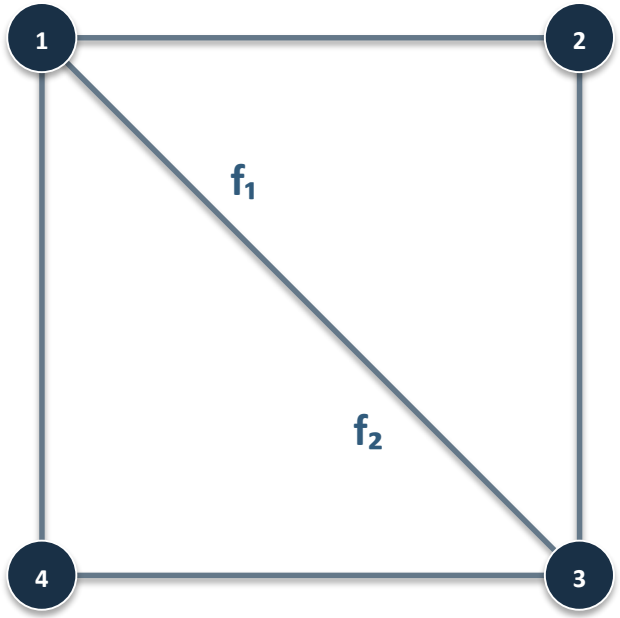


A crossing-free embedding

Planar = admits a crossing-free embedding | Plane = a chosen crossing-free embedding

Faces and Euler’s Formula

A topological invariant for connected plane graphs



f_3 (exterior)

$$V - E + F = 2$$

Example shown: V=4, E=5, F=3 → 4−5+3=2.

Every edge contributes twice to face-boundary degree:

$$\Sigma \deg(f) = 2E$$

This face-count relation is the bridge from topology to useful edge bounds.

Why Euler’s Formula Holds

Use the spanning-tree viewpoint rather than memorizing the equation

Start with a spanning tree — **Add a non-tree edge** — **One new face appears**

Tree: E = V−1, F = 1

E increases by 1

F increases by 1

V − E + F remains unchanged

For the spanning tree: V−(V−1)+1 = 2.

Each added edge creates exactly one additional face in a plane embedding.

Therefore the invariant remains 2 for the connected plane graph.

A Numerical Obstruction to Planarity

Derive the bound instead of treating it as a formula

Simple planar graph, V ≥ 3

Every face has boundary length ≥ 3

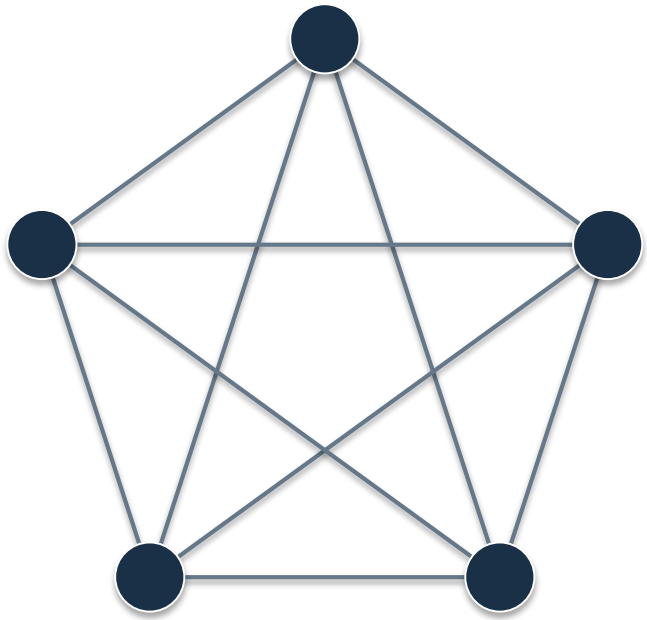
$$3F \leq 2E$$

Euler: $F = 2 - V + E$

$$3(2-V+E) \leq 2E$$

$$E \leq 3V - 6$$

Test on K_5



V=5, E=10; but 3V−6=9 → 10>9 → non-planar

Bipartite Planar Graphs Give a Stronger Bound

No odd cycles means no triangular faces

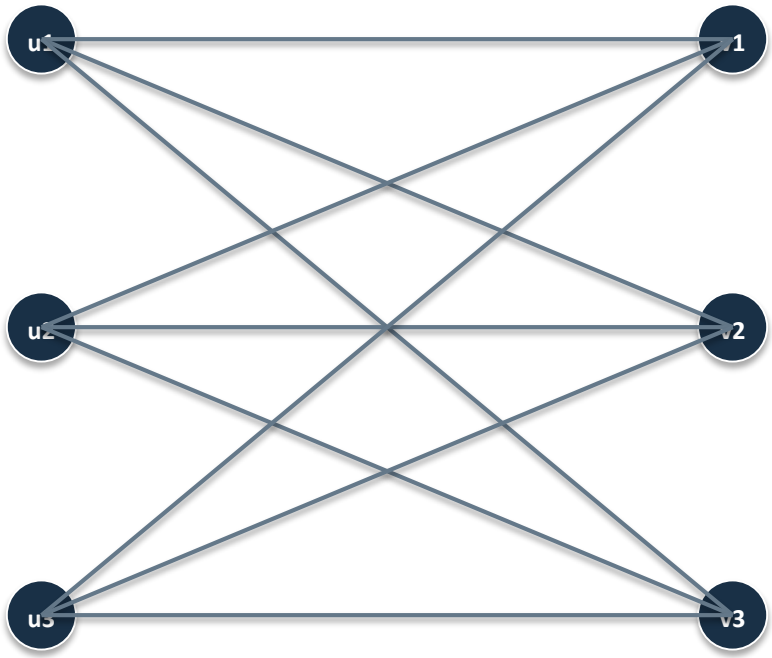
Bipartite ⇒ every face boundary has length ≥ 4

$$4F \leq 2E$$

With Euler's formula:

$$E \leq 2V - 4$$

$K_{3,3}$



V=6, E=9; but 2V−4=8 → non-planar

The Edge Bounds Are Necessary — Not Sufficient

Passing the inequality does not certify planarity

If planar ⇒ **$E \le 3V-6$**

BUT

$E \le 3V-6$ ⇏ **planar**

Edge-count tests detect some non-planar graphs, not all of them.

A graph can satisfy the bound and still contain a structural obstruction.

So the next question is not merely “How many edges?” but:

Does the graph contain the essential structure of K_5 or $K_{3,3}$?

Subdivision and Homeomorphism

Degree-2 vertices can hide a familiar non-planar structure



Original edge **Insert a degree-2 vertex**

Subdivision replaces an edge uv by a path u–x–v.

Suppressing a degree-2 vertex reverses this operation.

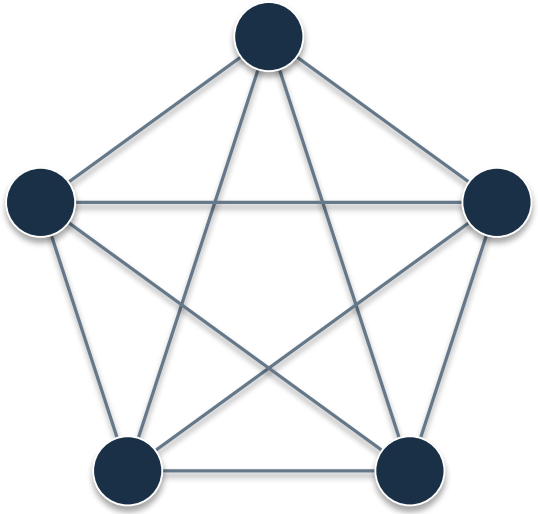
Two graphs are homeomorphic if they can be obtained from the same graph by subdivisions.

Planarity is preserved under subdivision.

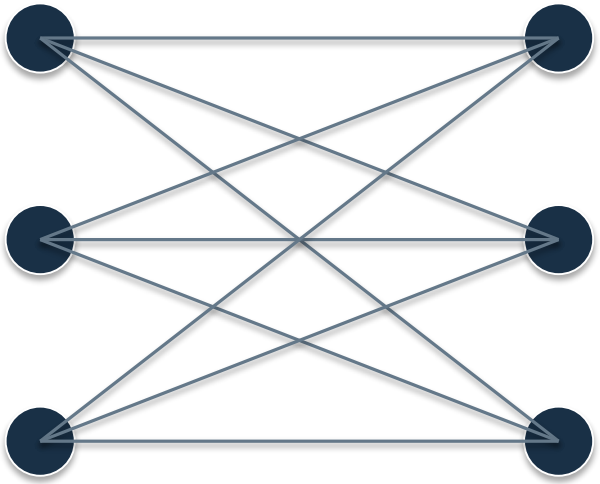
Kuratowski's Theorem: Structural Obstruction

A complete characterization of finite planar graphs

A graph is planar iff it contains no subdivision of K_5 or $K_{3,3}$.



K_5

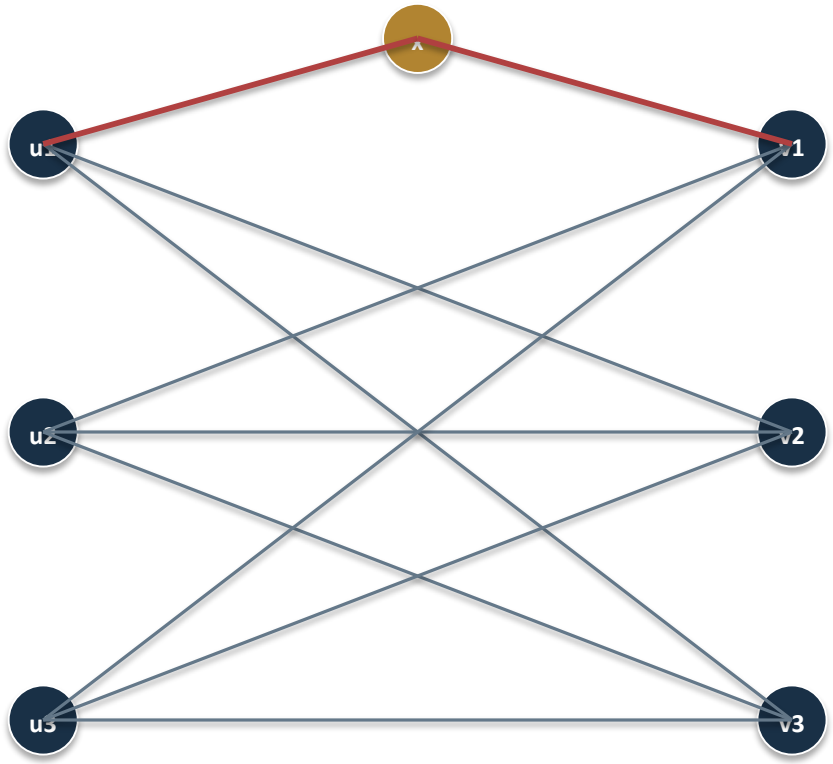


$K_{3,3}$

Think structurally: long paths of degree-2 vertices may simply be subdivided edges.

Worked Structural Test: Reveal the Hidden $K_{3,3}$

Suppress degree-2 vertices to expose the obstruction



Subdivided $K_{3,3}$

Vertex x has degree 2 and lies on the path $u_1–x–v_1$.

Suppress x: replace $u_1–x–v_1$ by edge u_1v_1.

The resulting graph is exactly $K_{3,3}$.

Therefore the original graph contains a subdivision of $K_{3,3}$ and is non-planar.

Structural certificate: subdivision of $K_{3,3}$

Choosing the Right Planarity Tool

Use the cheapest decisive argument first

1. Try to find / recognize an embedding
2. Check Euler-based edge bounds
3. If bipartite, use $E \leq 2V-4$
4. Bounds inconclusive? Search structurally

Look for a subdivision of K_5 or $K_{3,3}$

Numerical tests are fast but incomplete.

Kuratowski gives a structural characterization.

A valid proof should state why the chosen test applies.

Concept Check

Decide what can — and cannot — be concluded

A. Simple graph: $V=8$, $E=19$. Can it be planar? Justify.

B. Simple bipartite graph: $V=10$, $E=17$. Can it be planar? Justify.

C. A graph satisfies $E \leq 3V-6$. Is that enough to conclude it is planar?

D. A graph becomes $K_{3,3}$ after suppressing several degree-2 vertices. What follows?

Key distinction: edge-count obstruction vs. structural obstruction

Next: deeper planarity reasoning, dual graphs, and planarity detection.